LATEX

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Introduction

- **LaTeX** is a tool for creating professional-looking documents, especially useful for scientific and mathematical content.
- Unlike Microsoft Word, where you directly see and edit the formatted text, in LaTeX, you write your document as a plain text file with special commands.
- A program called a **TeX engine** processes this file to create a polished PDF. This way, you can focus on your writing, and LaTeX handles the formatting.

- A **TeX engine** is the software component that processes a document written in TeX or LaTeX and produces a formatted output, typically in the form of a PDF or DVI (DeVice Independent file format).

Introduction

- LATEX is a tool for typesetting professional-looking documents.
- However, LaTeX's mode of operation is quite different to many other document-production applications you may have used, such as Microsoft Word those "WYSIWYG" tools provide users with an interactive page into which they type and edit their text and apply various forms of styling.

- LaTeX is a typesetting system widely used for producing scientific and mathematical documents due to its powerful handling of formulas and bibliographies.
- LaTeX works very differently: instead, your document is a plain text file interspersed with LaTeX *commands* used to express the desired (typeset) results.

- To produce a visible, typeset document, your LaTeX file is processed by a piece of software called a *TeX engine* which uses the commands embedded in your text file to guide and control the typesetting process, converting the LaTeX commands and document text into a professionally typeset PDF file.
- This means you only need to focus on the *content* of your document and the computer, via LaTeX commands and the TeX engine, will take care of the *visual appearance* (formatting).

- LaTeX is a powerful tool for anyone needing to produce high-quality written content with complex layouts, especially in academia and research. Its ability to separate content from formatting allows authors to focus on writing while ensuring the final document looks polished and professional.

Why LaTeX

- **High-Quality Typesetting:**
  - Perfect for documents with complex math and scientific content.
- **Separation of Content and Style:**
  - Focus on writing; LaTeX handles the formatting.
- **Automatic Numbering and Cross-Referencing:**
  - Easily manage and reference sections, figures, tables, and equations.
- **Bibliographies and Citations:**
  - Efficiently manage references using tools like BibTeX.
- **Customization:**
  - Extend functionality with packages for graphics, algorithms, etc.
- **Document Classes:**
  - Use templates for articles, reports, books, and more.

Why Latex

- **High-Quality Typesetting:**
- Produces documents with professional-quality typesetting, particularly for documents containing
- complex mathematical equations and scientific content.
- **Separation of Content and Style:**
- Authors focus on writing content, while LaTeX takes care of the formatting.
- This separation ensures consistent styling throughout the document.
- **Automatic Numbering and Cross-Referencing:**
- Handles automatic numbering of sections, figures, tables, and equations, and allows
- easy cross-referencing within the document.

- **Bibliographies and Citations:**
- Manages bibliographies and citations efficiently using packages like BibTeX,
- which is particularly useful for academic papers.
- **Customization:**
- Highly customizable through a variety of packages that extend its functionality
- (e.g., for graphics, algorithms, or presentation slides).
- **Document Classes:**
- Supports different document classes (e.g., articles, reports, books)
- that cater to various types of publications.

Typesetting

- *the activity of arranging printed text and images on the page when preparing a book*, newspaper

- What is Latex
- When and Why People use
- What can you do in Latex
- Some Sample Document

Overview

- Leslie Lamport
- Document preparation system
- An alternative to text editing software
- Free software

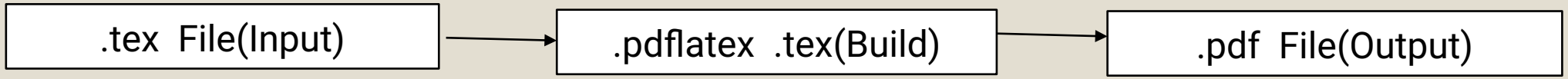
- It is not WYSIWYG
- Concentrate on content and not formatting
- Write content and complex mathematical formulae with ease

What can be done

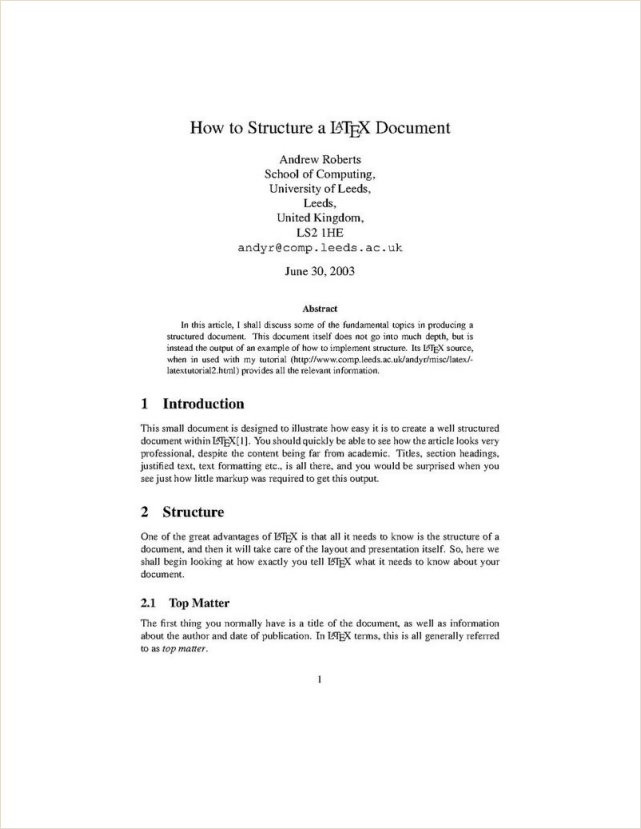
- Write articles,books, scientific journal, and papers
- Design poster, banner, resume
- Mostly used by writers and engineer

Process of writing

- Write content using any text editor:
  - Notepad
  - Gedit
  - Latex
- Save it as .tex
- Build
- Output: pdf file



Sample



How to Structure a LaTeX Document

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Abstract

In this article, I shall discuss some of the fundamental topics in producing a structured document. This document itself does not go into much depth, but is instead the output of an example of how to implement structure. Its LaTeX source, when in used with my tutorial (http://www.comp.leeds.ac.uk/andyr/misc/latex/-latextutorial2.html) provides all the relevant information.

1 Introduction

This small document is designed to illustrate how easy it is to create a well structured document within LaTeX[1]. You should quickly be able to see how the article looks very professional, despite the content being far from academic. Titles, section headings, justified text, text formatting etc., is all there, and you would be surprised when you see just how little markup was required to get this output.

2 Structure

One of the great advantages of LaTeX is that all it needs to know is the structure of a document, and then it will take care of the layout and presentation itself. So, here we shall begin looking at how exactly you tell LaTeX what it needs to know about your document.

2.1 Top Matter

The first thing you normally have is a title of the document, as well as information about the author and date of publication. In LaTeX terms, this is all generally referred to as *top matter*.

1

Install Latex

- Get introduce to different IDEs
- Get introduced Web based editor

Installation for windows

- Download Miktex

Installation ubuntu

- sudo apt-get install texlive-full

Latex IDE

- Texstudio
- Kile
- TeXworks

Web based Editor

- Overleaf
- LaTeX Base
- Papeeria
- Authorea

Editor Feature

- Auto complete of latex commands
- Parenthesis matching
- Collapsible section
- Inverse search
- No. of collaborator
- Public project
- Private project
- History
- Comments

Create First LaTeX Document

- Define Document class
- Provide document information
- Create your first latex document

Document class

- Type of document
- Use of appropriate class
- Syntax
  - \documentclass{article}

Document Information

- Title
  - \title {Creating first document in \LaTeX}
- Author
  - \author{Sufal Das}
- Date
  - \date {\today}

Document

- Write content
- \begin{document}

  ….
- \end{document}

Example

- \documentclass{article}
- \begin{document}

  Hello this is my first document in LaTeX
- \end{document}

- Organize the content into different sections
- Insert Table of contents

- Document class
  - article
  - books
  - report
- \documetclass{article}

Including title, author and date information

- \documentclass{article}
  - \title {Creating first document in \LaTeX}
  - \author{Sufal Das}
  - \date {\today}
- \begin{document}
  - \maketitle Hello this is my first document in LaTeX
- \end{document}

Sections

- Divide/classify content in document
- Associate with document class
- Has a level number

Section level

- -1. \part{…} : article, report,book
- 0. \chapter{…} : report,book
- 1. \section {…}
- 2. \subsection {…}
- 3. \subsubsection {…}
- 4. \paragraph {…}
- 5. \subparagraph {…}

- Note: 1- 5 I all document class

- \section{ Introduction}
- \subsection{Purpose}
- \subsection{Scope}

- \section {Overall Description}

- \documentclass{article}
  - \title {Creating first document in \LaTeX}
  - \author{Sufal Das}
  - \date {\today}
- \begin{document}
  - \maketitle

- \section{ Introduction}
- \subsection{Purpose}
- \subsection{Scope}

- \section {Overall Description}

\end{document}

Create a document in latex write about Meghalaya

- Comment - % (percentage symbol)
- Next - \\ (next line)

Table of Content

- \tableofcontent

- Display alias
  - \section[alias]{…}

- \documentclass{book}
  - \title {Creating first document in \LaTeX}
  - \author{Sufal Das}
  - \date {\today}
- \begin{document}
  - \maketitle
  - \tableofcontent

  - \chapter{overview}
- \section{ Introduction}
- \paragraph{}
- \subsection{Purpose}
- \subsection{Scope}

- \section {Overall Description}

\end{document}

Paper Size and Margin

- Uses packages
- Define different paper sizes
- Set margins

Using Packages

- File containing extra Latex command
- .sty file
- Styling new features
- \usepackage[Parameter] {PackageName}
- Mentioned before document begins

Paper size

- A series : A0,A1,A2,A3,A4….
- B series : B0,B1,B2,B3,B4…
- Wikipedia paper size
- \usepackage[a5paper] {geometry}

Margin

- Left,right,top
- \usepackage[margin=1.5] {geometry}
- \usepackage[a5paper] {geometry}
- \usepackage[a5paper, margin=1in]{geometry}
- \usepackage[a5paper, top=1in,left=2 in,right=2in,bottom=1in] {geometry}

Inner,outer/ odd ,even page

- \usepackage[a5paper, inner=2in,outer=1in]{geometry}

- **Customize header and footer**
- **Change page numbering style**
- **Reset page number**

Page Style

- Display header and footer
- Styles
  - Empty : no header nor footer
  - Plain : Footer contains the page number
  - Headings : Header contains section name and page number

- Entire Document
- \pagestyle{<style>}

- Current Page
- \thispagestyle{<style>}

- \pagestyle{empty}
- \thispagestyle{headings}

Reset Page number

- \setcounter{page}{<Number>}

Page Numering style

- Arabic: 1,2,3
- roman : I,ii,iii
- Roman : I,II,III
- alph : a,b,c
- Alph : A, B, C
- \pagenumbering{<style>}

- E.g
- \pagenumbering{roman}

- Set orientation
- Add page breaks
- Set footnotes

Orientation

- Direction of a document
- Portraits (vertical) - height is larger than the width
- Landscape (horizontal)

Landscape

- \usepackage[landscape] {geometry}

- \documentclass{article}
- \usepackage{lipsum}
- \usepackage[landscape] {geometry}

- \begin{document}
  - \lipsum[1-10]
- \end{document}

Specific part

- Package
  - \usepackage{lscape}
  - **Rotates content** by 90 degrees

  - \usepackage{pdflscape}
  - **Rotates page** by 90 degrees

- \begin{landscape}

- \end{landscape}

lscape package

- \documentclass{article}
- \usepackage{lipsum}
- \usepackage{lscape}

- \begin{document}
  - \section{Title}
  - \lipsum[1-2]

  - \begin {landscape}
  - \section{Second Title}
  - \lipsum[1-2]
  - \end{landscape}

  - \lipsum[1-10]

- \end{document}

pdflscape package

- \documentclass{article}
- \usepackage{lipsum}
- \usepackage{pdflscape}

- \begin{document}
  - \section{Title}
  - \lipsum[1-2]

  - \begin {landscape}
  - \section{Second Title}
  - \lipsum[1-2]
  - \end{landscape}

  - \lipsum[1-10]

- \end{document}

Page breaks

- \pagebreak
- \newpage

\pagebreak

- \documentclass{article}
- \usepackage{lipsum}
- \usepackage{pdflscape}

- \begin{document}
  - \section{Title}
  - \lipsum[1-2]
  - **\pagebreak**

  - \section{section in between}
  - \lipsum[1-2]

  - \begin {landscape}
  - \section{Second Title}
  - \lipsum[1-2]
  - \end{landscape}

  - \lipsum[1-10]

- \end{document}

Footnotes

- More information about the content
- Written at the bottom
- \footnote{text to be display as footnote}

Footnotes

- \documentclass{article}
- \usepackage{lipsum}
- \usepackage{pdflscape}

- \begin{document}
  - \section{Title}
  - \lipsum[1-2]
  - LaTeX is widely used in academia for the communication and publication **\footnote**{refer to wikipedia for more info}what else do you need **\footnote**{contact for any help}
  - \pagebreak

  - \section{section in between}
  - \lipsum[1-2]

  - \begin {landscape}
  - \section{Second Title}
  - \lipsum[1-2]
  - \end{landscape}

  - \lipsum[1-10]

- \end{document}

Overview

- Set one column document
- 2 column
  - Research papers
  - Books
- Multi Columns
  - Megazine
  - Newspaper

Two column

- \documentclass[twocolumn]{…}

- E.g
- \documentclass[twocolumn]{article}
- \usepackage{lipsum}

- \begin {document}
  - \lipsum[1-10]
- \end{document}

Multi Columns

- Package : multicol
- Syntax
- \begin {multicols} {<No. of columns>}
- ….
- \end {multicols}

- E.g
- \documentclass{article}
- \usepackage{lipsum}
- \usepackage{multicol}

- \begin {document}

- \begin {multicols}{3}
- \lipsum[1-10]
- \end{multicols}

- \end{document}

Add space between column

- \setlength{\columnsep}{20pt}

- E.g
- \documentclass{article}
- \usepackage{lipsum}
- \usepackage{multicol}

- \begin {document}
- **\setlength{\columnsep}{20pt}**
- \begin {multicols}{3}
- \lipsum[1-10]
- \end{multicols}

- \end{document}

Dividing line

- \setlength{\columnseprule}{3pt}

- E.g
- \documentclass{article}
- \usepackage{lipsum}
- \usepackage{multicol}

- \begin {document}
- \setlength{\columnsep}{20pt}
- **\setlength{\columnseprule}{3pt}**
- \begin {multicols}{3}
- \lipsum[1-10]
- \end{multicols}

- \end{document}

\columnbreak

- \documentclass{article}
- \usepackage{lipsum}
- \usepackage{multicol}
- \begin{document}
  - \setlength{\columnsep}{20pt}
  - \setlength{\columnseprule}{3pt}

- \begin{multicols}{2}
- Row, row, row your boat,Gently down the stream,Merrily, merrily, merrily, merrilyLife is but a dream.

- **\columnbreak**
- Jack and Jill went up the hillTo fetch a pail of water.Jack fell down and broke his crown,And Jill came tumbling after.

- \end{multicols}
- \end{document}

Formatting Multicolumns

- Column spacing
  - Spacing between the columns
  - \setlength {\columnsep}{<X>pt}

- Column separator
  - Dividing columns with a line
  - \setlength {\columnseprule} {<X>pt}

- Column Break
  - Force text into next column
  - \columnbreak

Read and solve Error

Overview

- Building .tex file
- Display error message if any issue
- Missing or mismatch in curly brace
- Environment not ended
- Package not included
- Style file not found

Structure of Error Message

- Structure
  - ! <Error Message>
  - 1. <Line Number>

Undefined control sequence

- Chapter in article
- \documentclass{article}
- \begin{document}
- \chapter{title}

- Error message
- Undefined control sequence. \chapter

- \chapter are not define in article class

- \documentclass{article}
- \begin{document}
- \begin{landscape}
  - Content..
- \end{landscape}
- \end{document}

- Solution: \usepackage{lscape} need to be declared,package is not define

- Now try add s in \usepackage{lscapes}
- Error will be not found

- Try remove one closing curly braces
- It will end on that line itself

Package not included

- Lscape package not written
- \begin{landscape}
- …
- \end{landscape}

- ERROR Message
- !LaTex Error ： Environment landscape undefined.

- Latex
- \usepackage{lscapes} instead of \usepackage{lscape}

- ERROR Message
- !LaTex Error : File ‘lscapes.sty’ not found

Topic 3

- Formatting Text
  - Increase/decrease font size
  - Use different font size

- Unlike other WYSIWYG editors, the content is written in plain text along with appropriate commands, thus allowing the user to concentrate on the content rather than the aesthetics (the way it looks).
- The teX typesetting program which LaTex uses, was design such that anyone can create good quality material with less efforts.

- Text Bold - \textbf{<text>}
- Emphasize (italic) - \emph{< text >}
- Underline - \underline{< text >}

Font Style

- \textbf{..} **Text**
- \textit{..} *text*
- \emph{..} *Text*
- \underline{..} <u>Text</u>
- \textsc{..} TEXT
- \texttrm{..} Text
- \textsf{..} Text
- \texttt{..} Text
- ^{..} Text^{Text}
- \textsubscript{..} Text$_{\text{Text}}$

Font size

- \tiny The smallest font
- \scriptsize Font size for scripts
- \footnotesize Footnote font size
- \small Small Font size
- \normalsize The default font size
- \large large font size
- \Large Large Font Size
- \LARGE LARGE Font Size
- \huge huge font size
- \Huge The largest font size

Noted

- \Large TeX - make large as well as the following TeX
- \Large {TeX} – same make large as well as the following TeX
- {\Large TeX} – put the beginning curly braces before latex command \large

- \documentclass{article}
- \usepackage{lipsum}
- \begin{document}
- \tiny
- \lipsum[1-2]

- \normalsize
- \lipsum[3-4]
- \end{document}

Color text

- Assign a color to your text
- Assign a color to your page

overview

- Basic color
  - Red
  - Blue
  - Green
  - White
  - Yellow etc

  - Package: color

- More color
  - Package :xolor
  - Parameter : dvipsnames

  - https://en.wikibooks.org/wiki/LaTeX/Colors

Text Color

- Single Color
- \textcolor{blue}{Text}
- My text is coloured blue

- Mixing with white
- \textcolor{red!70} {text}
- A mixture of 70% red and 30% white

- Mixture of two colors
- \textcolor{red!70!blue} {}
- Mixture of red and blue

- \documentclass{article}
- \usepackage[dvipsnames]{xcolor}
- \begin{document}
- \pagecolor{Goldenrod!50}
- \textcolor{red} {This is red colour}
- \textcolor{blue} {This is blue colour}

- \textcolor{blue!70!red} {This is blue and red colour}

- \end{document}

Page color

- \pagecolor{red!50}

Align Text

- Align your text (left,right,center)

Alignment

- Default: Justify %\usepackage{ragged2e}
- Left Align
- \begin{flushleft} ……\end{flushleft}

- Center Align
- \begin{center} ……\end{center}

- Right Align
- \begin{flushright} ……\end{flushright}

- \documentclass{article}
- \usepackage{lipsum}
- \usepackage{ragged2e}
- \begin{document}

- \section {one}
- \begin{flushleft}
- \lipsum[2-2]
- \end{flushleft}

- \section {Two}
- \begin{center}
- \lipsum[2-2]
- \end{center}

- \section {Three}
- \begin{flushright}
- \lipsum[2-2]
- \end{flushright}

- \pagebreak
- \centering
- \lipsum[1-1]

- \flushleft
- \lipsum[2-2]

- \flushright
- \lipsum[3-3]

- \justify %use package \usepackage{ragged2e} default format
- \lipsum[4-4]

- \end{document}

ragged2e

- Package : ragged2e
- Switch back: \justify

Spacing

- Add single , one and a half, and double line spacing
- Insert vertical and horizontal spacing

Fixed spacing

- Package : setspace
- Entire document
  - \singlespacing
  - \onehalfspacing
  - \doublespacing
- Certain part
- \begin{singlespace | onehalfspace | doublespace}
- …
- \end{}

- \documentclass{article}
- \usepackage{lipsum}
- \usepackage[margin=1in]{geometry}
- \usepackage{setspace}

- \begin{document}
- \doublespacing
- \lipsum[1-2]

- \lipsum[1-2]

- \end{document}

- \documentclass{article}
- \usepackage{lipsum}
- \usepackage[margin=1in]{geometry}
- \usepackage{setspace}

- \begin{document}
- \begin{doublespace}
- \lipsum[1-2]
- \end{doublespace}
- \lipsum[1-2]

- \end{document}

Vertical Spacing

- \vspace{Xpt}
- \smallskip
- \medskip
- \bigskip
- \vfill

- \documentclass{article}
- \usepackage{lipsum}
- \usepackage[margin=1in]{geometry}
- \usepackage{setspace}

- \begin{document}
- \begin{doublespace}
- \lipsum[1-2]
- \end{doublespace}
- \lipsum[1-2]

- \pagebreak
- \section {smallskip}
- \smallskip
- \lipsum[1-2]

- \section {medskip}
- \medskip
- \lipsum[1-2]

- \section {bigskip}
- \bigskip
- \lipsum[1-2]

- \vspace{pt}
- \lipsum[1-2]

- \end{document}

- \pagebreak

- Lipsum[2-2]
- \vfill
- Lipsum[2-2]
- \vfill
- Lipsum[2-2]

Horizontal Spacing

- \ single space
- \, more
- \hspace{Xpt}
- \hfill

- Mostly use while typesetting mathematics

- \pagebreak
- \noindent
- \LaTeX \ Document \\
- \LaTeX \, Document \\
- \LaTeX \hfill Document \\

Bullets and Numering

- Create ordered list
- Create unordered list
- Create Description list
- Change the bullet style
- Change the numering

Ordered list

- Numbered list
- \begin{enumerate}

  \item Red

  \item Blue

  \item Green
- \end{enumerate}

Unordered list

- Bullet list
- \begin{itemize}

    \item Mouse

    \item Keyboard

    \item Monitor
- \end{itemize}

Description List

- Labeled list
- \begin{description}

    \item [Input Devices] : mouse, keyboard

    \item [Output Device] : monitor , printer
- \end{description}

Nested List

```
◦ \section{Numbered list}
◦ \begin{enumerate}
◦ \item Red
        \begin{itemize}
            \item Blood Red
            \item Brick Red
        \end{itemize}
◦ \item Blue
        \begin{enumerate}
            \item Navy Blue
            \item Sky Blue
        \end{enumerate}
◦  \item Green
◦ \end{enumerate}
```

- List inside another list
- Combination of un-ordered and order

- \begin{itemize}

```
\item Input Devices
        \begin{enumerate}
                \item keyboard
        \end{enumerate}
\item Output Devices
        \begin{itemize}
                \item printer
        \end{itemize}
\end{itemize}
```

Change Numering in Enumerate

- Package : enumerate

- Style : (i, ii) ,(I, II), (a,b), (A,B), (1,2)
- Syntax
- \begin{enumerate} [<style>]

- Example
- \begin{enumerate} [(A)]
- \begin{enumerate} [i.]

Change Bullet Style

- Package : amssymb (ams means American Mathematical Society) symbol

- Style:
- \blacksquare, \square, \bullet, \cdot, \diamond , - , * , \circ

- Level : i ,ii,iii,iv

- Command
- \renewcommand{\labelitem<item>}{$<Style>$}

- Example
- \renewcommand{\labelitemi<item>}{$<diamond>$}
- \renewcommand{\labelitemii<item>}{$<diamond>$}

- \documentclass{article}
- \usepackage{lipsum}
- \usepackage{enumerate}
- \usepackage{amssymb}
- \begin{document}

- \renewcommand{\labelitemi}{$\diamond$}
- \renewcommand{\labelitemii}{$\blacksquare$}

```
\section{Bullet list}
\begin{itemize}
    \item Mouse
    \begin{itemize}
        \item blue color
        \item black color
    \end{itemize}

    \item Keyboard
    \item Monitor
\end{itemize}

\end{document}
```

Creating Tables

- Creating a table using tabular environment
- Align content of column
- Specify column width

Overview

- Tabular environment

- \begin{tabular} {cols}
- …
- \end{tabular}

- Cols
  - l (left)
  - r (right)
  - c (center)
  - p{Xcm} : left align (we can adjust the size of the column)

- Column : &
- Row : \\

Basic Table

- No. Item Price
- 1 Pen 10
- 2 Pencil 10

```
\documentclass{article}
\begin{document}
\begin{tabular}{clr}
No.     & Item    & Price          \\
1       & Pen     & 10     \\
2       & Pencil  & 10     \\
\end{tabular}
\end{document}
```

- Now try to add more row

```
\begin{tabular}{clr}
No.     & Item    & Price   \\
1       & Pen     & 10      \\
2       & Pencil  & 10      \\
3       &Books    &1567
\end{tabular}
```

Now try to add more column

- \begin{tabular}{clrp{2cm}}
- No. & Item & Price &Quantity\\
- 1 & Pen & 10 &3 \\
- 2 & Pencil & 10 &5 \\
- 3 &Books &1567 &1 \\
- \end{tabular}

Table Borders

- Vertical : |
- Horizontal : \hline
- Specific: \cline{i-j}

Table with Border

No.	Price	Item	Quantity
1	Pen	10	3
2	Pencil	10	5

```
\begin{tabular}{|c|l|l|r|}
\hline
◦ No.     & Item    & Price   &Quantity\\
◦ \hline
◦ 1       & Pen     & 10      &3        \\
◦ \hline
◦ 2       & Pencil  & 10      &5        \\
\end{tabular}
```

cline

No.	Price	Item	Quantity
1	Pen	10	3
2	Pencil		5

```
\begin{tabular}{|c|l|l|r|}
\hline
No.  & Item     & Price    &Quantity\\
\hline
1      & Pen      & 10&3    \\
\cline{1-2} \cline{4-4}
2      & Pencil   &     &5    \\
\hline
\end{tabular}
```

Dash border

- Package : arydshln (array dash line)
- Vertical : **:** (instead of pipe operator | replace with colon :)
- Horizontal: **\hdashline[size of dash / space between dashes]** (instead of \hline)
- Specific : cdashline {i-j} [size of dash / space between dashes]

- Example:
- \hdashline [1pt / 2pt]
- \cdashline{2-3} [1pt/ 2 pt]

Example : Dash Border

No.	Price	Item	Quantity
1	Pen	10	3
2	Pencil		5

```
\begin{tabular}{:c:l:l:r:}
\hdashline[3pt/5pt]
◦ No.      & Item     & Price    &Quantity\\
\hdashline[3pt/5pt]
◦ 1        & Pen      & 10       &3          \\
◦ \cdashline{1-2} [3pt/5pt]
◦  \cdashline{4-4} [3pt/5pt]
◦ 2        & Pencil   &          &5          \\
\hdashline[3pt/5pt]
\end{tabular}
```

Merging Rows and Columns

- Merge row

Merge row

- Package : multirow
- Syntax : \multirow {NumberofRows}{width}{..}

No.	Price	Item	Quantity
1	Pen	10	3
2	Pencil		5

```
\begin{tabular}{|c|l|l|r|}
\hline
No.     & Item    & Price   &Quantity\\
\hline
1       & Pen     & 10      &3        \\
\cline{1-2} \cline{4-4}
2       & Pencil  &         &5        \\
\hline
\end{tabular}
```

```
◦ \documentclass{article}
◦ \usepackage{multirow}
◦ \begin{document}
\begin{tabular}{|c|l|l|r|}
\hline
◦ No.      & Item     & Price    &Quantity\\
◦ \hline
◦ 1        & Pen      & \multirow{2}{*}{10}        &3        \\        %(* denote natural width)
◦ \cline{1-2} \cline{4-4}
◦ 2        & Pencil   &        &5        \\
◦ \hline
\end{tabular}
◦ \end{document}
```

Merge Columns

- \multicolumn {TotalColumns}{Alignment}{...}

stationary Items			
No.	Items	Quantity	Price

- \renewcommand{\arraystretch}{1.5}
- \begin{tabular} {|c|l|l|r|}
- \hline
- \multicolumn{4}{|c|} {Stationary Items} \\
- \hline
- No. &Item &Quantity &Price \\
- \hline

Large Table

- A large table
- Content not getting display
- Limitation : Manually split data

- \documentclass{article}
- \usepackage{longtable}
- \begin{document}
- \large
- \begin {**longtable**}{|l|l}
- \hline
- \multicolumn{2}{|c|} {Stationery Items} \\
- \hline
- No. &item \\
- \hline
- \endhead
- 1 & Pen \\
- \hline
- 2 & Pencil \\
- \hline

- \end{**longtable**}
- \end{document}

- Handle large data in a table

Long Table

- Package : longtable
- Environment
- \begin{longtable}{columns}
- …
- \end{longtable}

Table Environment

- Add captions
- Position Table
  - t: top
  - b : bottom
  - h : approximately here

- Environment
- \begin {table} [position] //note that position is optional

  your tabular code

  \caption{…}

\end{table}

Example

- \begin{table} [t]
- \begin{tabular}{l|l}
- No. & items \\
- 1 & Pen \\
- …
- \end{tabular}
- \caption{Stationery items]
- \end{table}

```
◦ \documentclass{article}
◦ \usepackage{lipsum}
◦ \begin{document}
◦ \tableofcontents
◦ \listoftables
◦ \section{one}
◦ \lipsum[1-2]

◦ \begin{table} [b]
◦ \centering
◦ \begin{tabular}{|l|l|}
◦ \hline
◦ No.      &item      \\
◦ \hline
◦ 1        &Pen       \\
◦ \hline
◦ 2        &Pencil    \\
◦ \hline
◦ 3        &scale     \\
◦ \hline
◦ \end{tabular}
```

- \caption{Stationery Items}
- \end{table}

- \section{Two}
- \lipsum[1-3]

- \begin{table} [h]
- \centering
- \begin{tabular}{|l|l|}
- \hline
- No. &item \\
- \hline
- 1 &Pen \\
- \hline
- 2 &Pencil \\
- \hline
- 3 &scale \\
- \hline
- \end{tabular}

- \caption{More Items}
- \end{table}

Include Graphics

- Include images and resize them

- Package : graphicx
- Supported formats : jpg,png,eps,pdf
- Resize
- Scale
- Trim/crop
- Rotate
- Border

Syntax

- \includegraphics [attributes]{filename.extn}

- Attribute to change size

  width

  height

Example

- \includegraphics[height=4cm,width=3cm]{filename.jpg}

- \documentclass{article}
- \usepackage{graphicx}
- \begin{document}
- \includegraphics[height=5cm,width=5cm]{image.jpg}
- \end{document}

Scale, Crop, Rotate images

Scale

- Resizing an image
- Scaled : \includegraphics[scale=0.5]{filename.jpg}

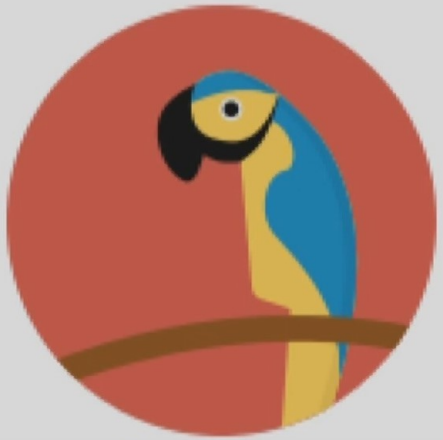


Figure: Original

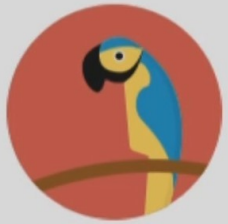


Figure: Scaled

Crop

- Cropped: \includegraphics[clip=true,trim= 15mm 5mm 7mm 3mm] {filename.jpg}
- Four values for trimming the image,in the order of left,bottom,right, and top

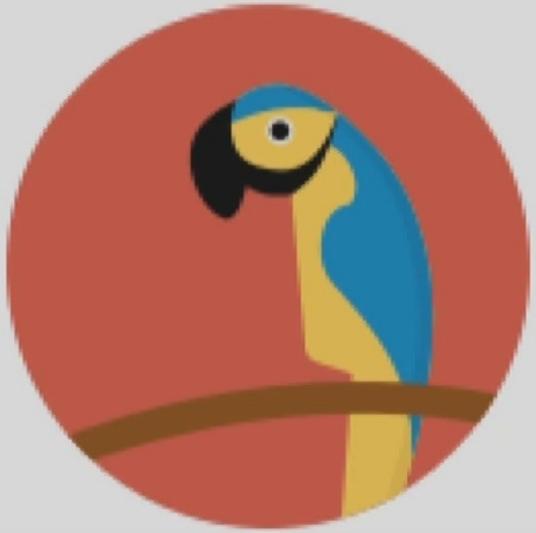


Figure: Original

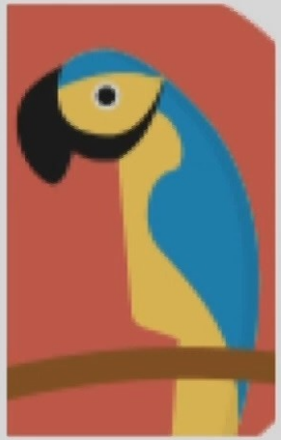


Figure: Cropped

Rotate

- Rotate : \includegraphics [angle =60] {filename.jpg}

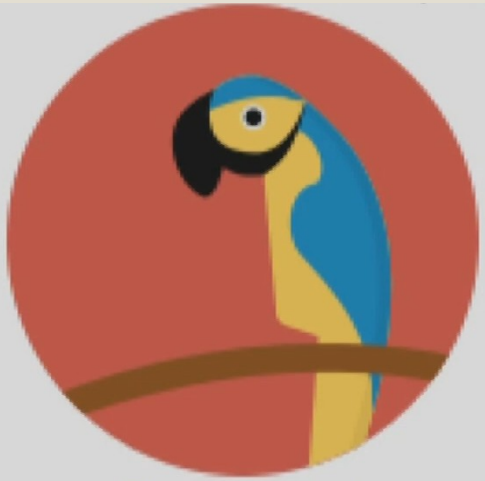


Figure: Original

Figure: Rotated

Example

- \documentclass{article}
- \usepackage{graphicx }
- \begin{document}

- \section{original image}
- \includegraphics{filename.jpg}

- \section{Rotate image}
- \includegraphics [angle =60] {filename.jpg}

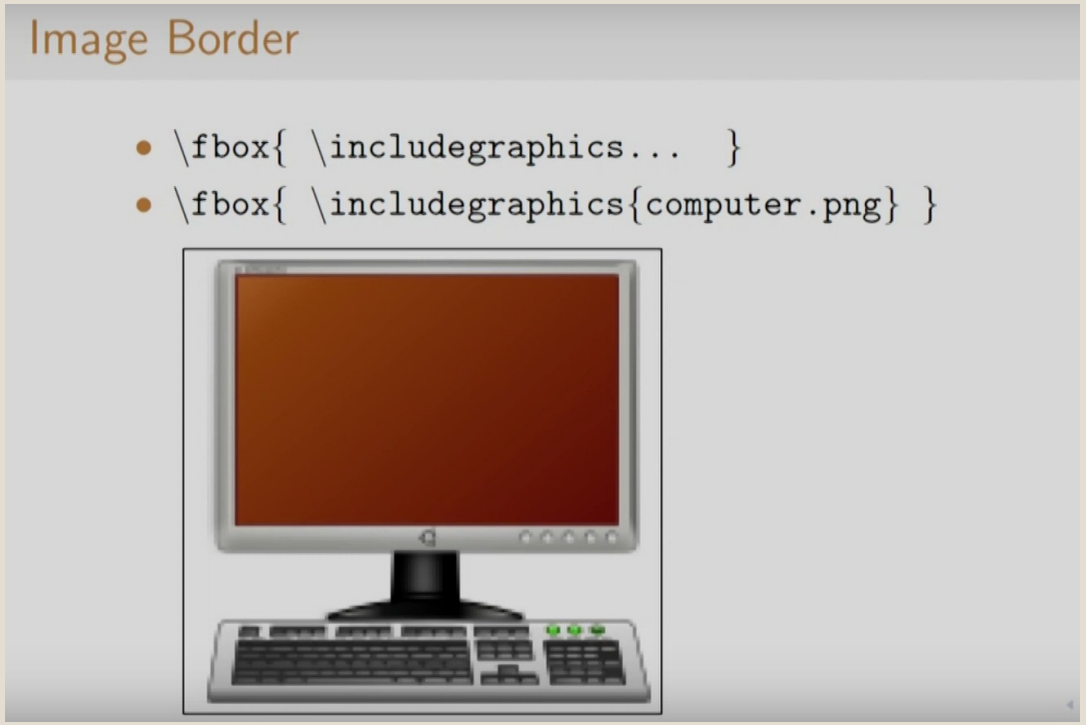
- \section{Crop image}
- \includegraphics [clip =true,trim=5mm 5mm 5mm 5mm] {filename.jpg}

- \section{Scale image}
- \includegraphics [scale=0.5] {filename.jpg}

- \end{document}

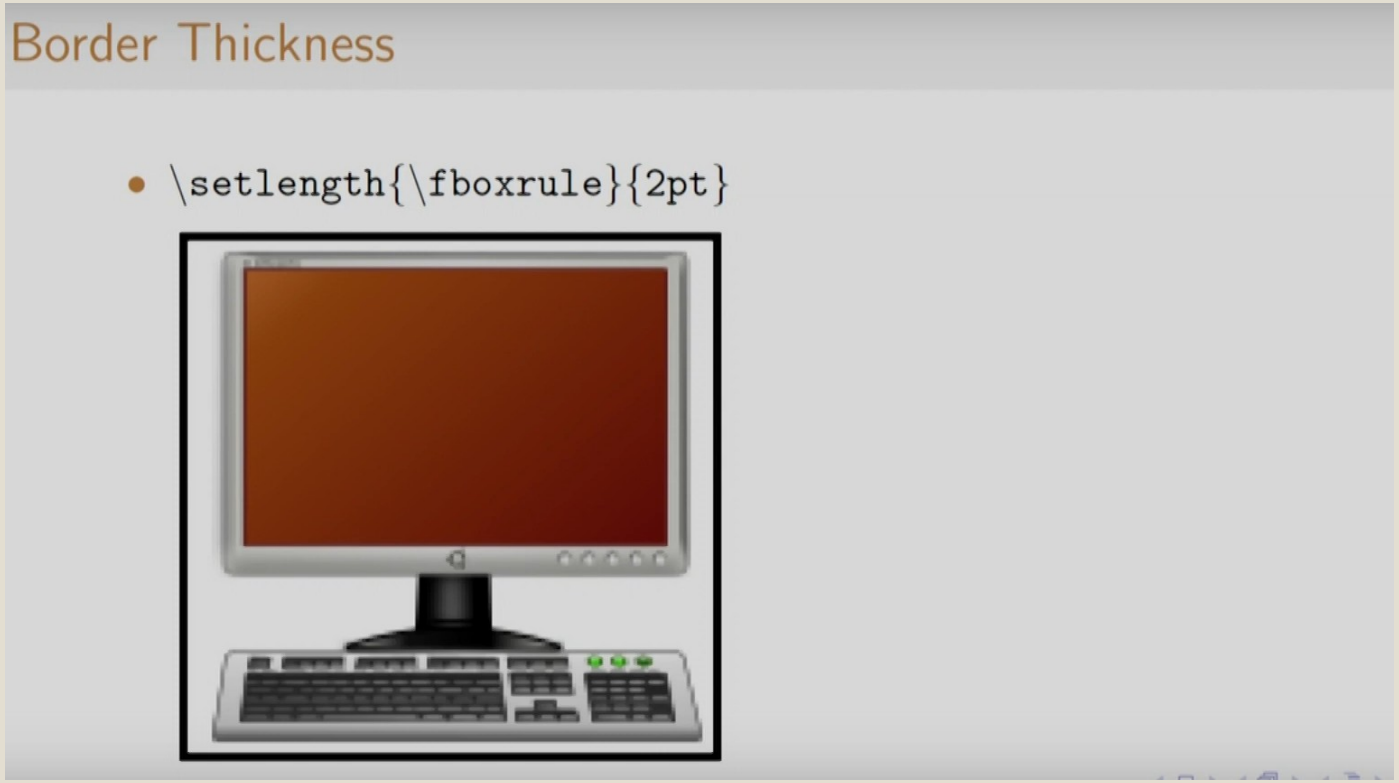
Border to image

- \fbox{\includegraphics…}
- \fbox{\includegraphics{image.jpg}}



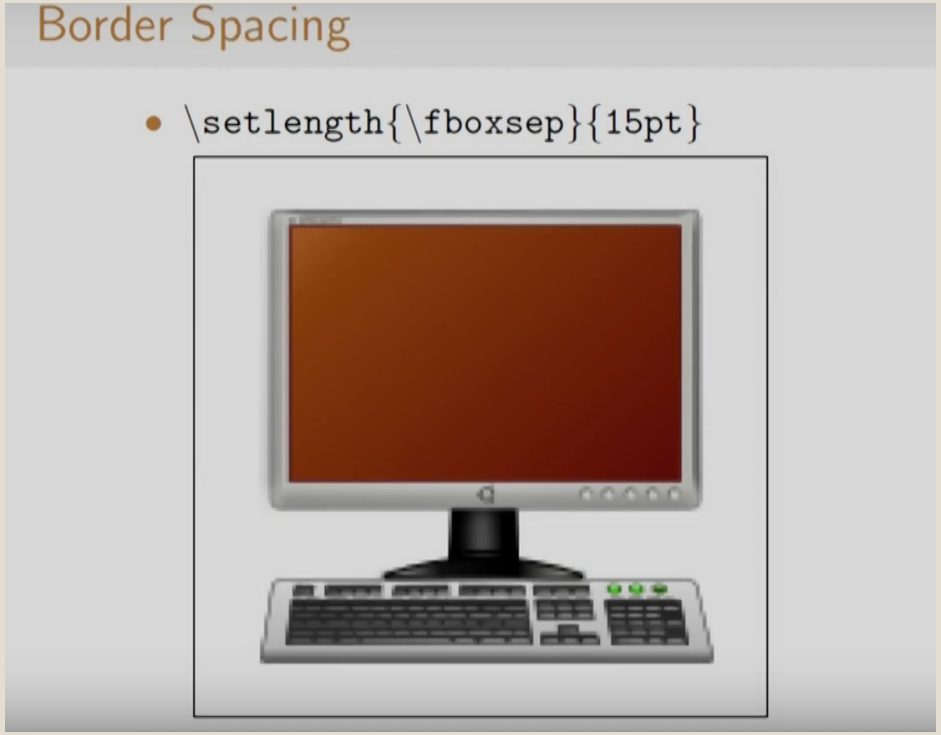
Border thickness

- \setlength{\fboxrule}{2pt}



Border Spacing

- \setlength {\fboxsep}{15pt}



Example

- \documentclass{article}
- \usepackage{graphicx }
- \begin{document}

- \setlength{\fboxrule}{2pt}
- \setlength {\fboxsep}{15pt}
- \fbox{\includegraphics{image.jpg}}

- \section{Border image}
- \fbox{\includegraphics{filename.jpg}}

- \end{document}

Figure and Sub Figure

- Specify caption for a figure
- Generate list of figure
- Include multiple image under one image

Figure Environment

- Environment
- \begin{figure}[position]
- \includegraphics[...]{...}
- \caption{...}
- \end{figure}

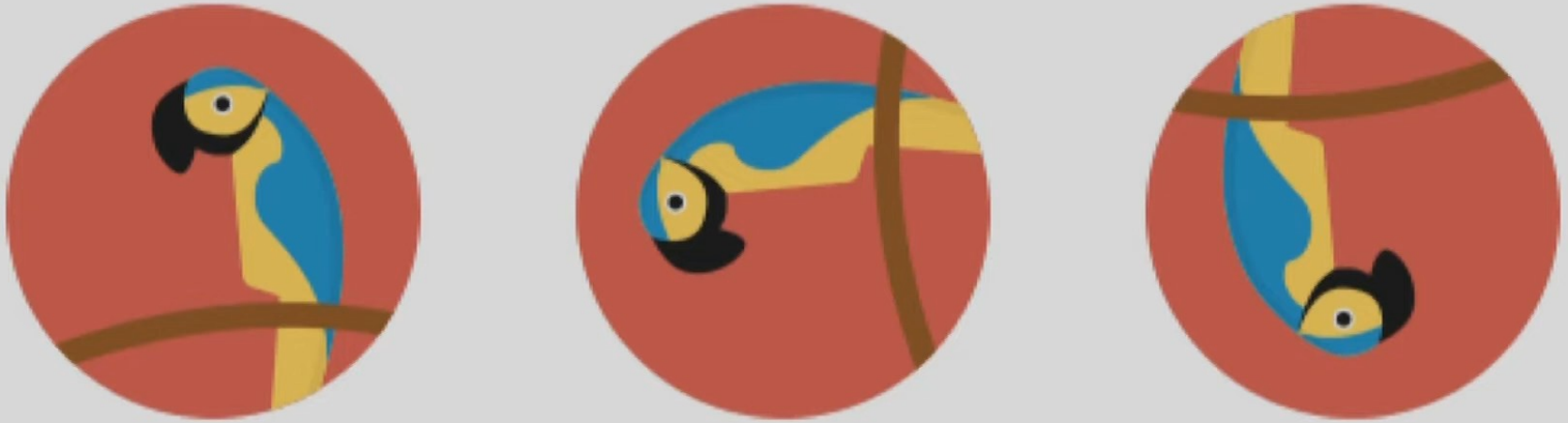
- List
- \listoffigures

Example

- \documentclass{article}
- \usepackage{graphicx}
- \begin{document}
- \begin{{figure}[h]
- \centering
- \includegraphics[width=1.5in]{image.jpg}
- \caption{Original image}
- \end{figure}
- \end{document}

Sub figure

Sub-Figure



(a) Original (b) Rotated: 90 degrees (c) Rotated: 180 degrees

Figure: Image of a Parrot

Sub figure

- Package: caption, subcaption

- Structure
- \begin{figure}[<position>]
- \begin{subfigure}{<width>}
- \includegraphics…
- \caption{…}
- \end{subfigure}
- …
- \caption{…}
- \end{figure}

Example

- \documentclass{article}
- \usepackage{graphicx}
- \usepackage{caption}
- \usepackage{subcaption}

- \begin{document}
- \begin{figure}
- \centering

- \begin{subfigure} {0.3\textwidth}
- \centering
- \includegraphics[width=\textwidth]{image.jpg}
- \caption{picture 1 }
- \end{subfigure}
- \hfill

- \begin{subfigure} {0.3\textwidth}
- \centering
- \includegraphics[width=\textwidth]{image.jpg}
- \caption{picture 2 }
- \end{subfigure}
- \hfill
- \begin{subfigure} {0.3\textwidth}
- \centering
- \includegraphics[width=\textwidth]{image.jpg}
- \caption{picture 3}
- \end{subfigure}

- \end{figure}
- \end{document}

Title Page

- title
- author
- date
- \maketitle